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Selective, Regularized, and Calibrated: Harnessing Vision Foundation Models for Cross-Domain Few-Shot Semantic Segmentation

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Abstract

Vision foundation models (VFMs) have achieved strong performance across various vision tasks. However, it still remains challenging to apply VFMs for cross-domain few-shot segmentation (CD-FSS), which segments objects of novel classes under domain shifts using only a few labeled exemplars. The challenge is mainly driven by two factors: (1) limited labeled exemplars per novel class relative to the scale of VFM pre-training, making the model prone to overfitting during retraining, and (2) target-domain shifts underrepresented during pre-training, inducing cross-domain inconsistency and layerwise sensitivity. To address these issues, we propose Hierarchical Exemplar Representation Adaptation (HERA), a three-stage select-regularize-calibrate VFM-based segmentation framework that learns effectively from limited labels and adapts to novel domains without source-data retraining. We first design Hierarchical Layer Selection (HLS) to adaptively identify the most informative VFM layer using a data-dependent Exemplar Transfer Risk (ETR) computed for each candidate layer. Then, Prior-Guided Regularization (PGR) regularizes interactions on the selected representation, yielding well-structured local signals for the subsequent stage. Furthermore, Pixelwise Adaptive Calibration (PAC) combines the selected representation with the refined interaction maps to calibrate pixelwise predictions, producing consistent masks. Together, these stages form a hierarchical select–regularize–calibrate pipeline that guides frozen VFM features in new domains while fine-tuning less than 2.7% of parameters at test time. Extensive experiments show that HERA surpasses the state-of-the-art by more than 4.1 mIoU across multiple CD-FSS benchmarks.

1. Introduction

Few-shot semantic segmentation (FSS) aims to generate pixel-level predictions for novel classes using only a few labeled support exemplars. Existing methods [12, 38, 53, 60] learn class-agnostic correspondences from constructed pairs of support and query images, which transfers knowledge from base to novel classes and yields strong in-domain generalization. However, when the deployment distribution differs from the training distribution, the learned correspondences and class prototypes degrade, leading to large performance drops. This motivates cross-domain few-shot segmentation (CD-FSS), which seeks to generalize to novel classes in unseen target domains under distribution shift, using only a few labeled support exemplars.

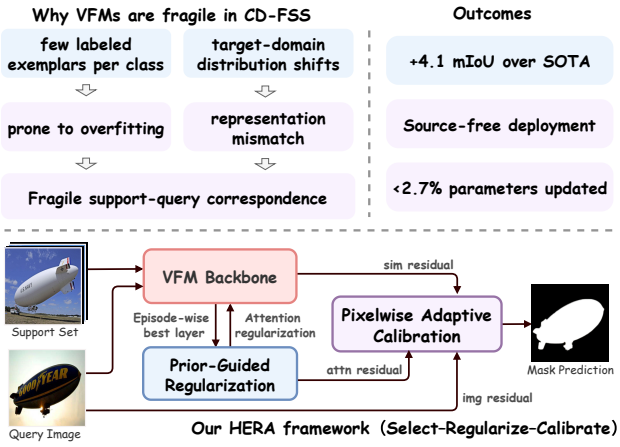


Figure 1. Scarce labels and target domain shift co-occur in CD-FSS, making VFMs fragile in deployment. Retraining tends to overfit and misalign representations, thereby destabilizing support–query correspondence. Our method HERA, a three-stage select-regularize-calibrate framework, adapts a frozen VFM at test time with only limited labeled exemplars and no source data, achieving state-of-the-art performance.

Existing CD-FSS methods [21, 39, 48] predominantly rely on CNN backbones, and typically extend in-domain FSS either by retraining on source data with domain-generalization techniques or by mining cross-image correspondences. Although effective in controlled settings, these approaches are costly and depend on source data. Their convolutional inductive biases limit long-range reasoning

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and robustness under distribution shift, leading to overfitting with sparse labels. Given these limitations, replacing CNN backbones with vision foundation models (VFMs) yields stronger and more transferable representations.

VFMs are pretrained vision backbones that provide transferable representations across recognition, segmentation, and multimodal tasks [7, 40, 45]. Despite these strengths, applying them to CD-FSS is challenging for two reasons. First, the few-shot regime offers only a handful of labeled exemplars per novel class, so retraining is prone to overfitting, source-dependent, and computationally prohibitive for large models. Second, distribution shifts place target domains outside pretraining coverage, so using frozen features or adapting all layers indiscriminately yields cross-domain inconsistency and pronounced layer sensitivity, which destabilize correspondence and pixel predictions. Empirically, VFM features exhibit sharp layerwise variation in transferability under shift [29, 43, 44]. Adapting a single stable layer reduces degrees of freedom and curbs overfitting. Notably, most recent VFM-based CD-FSS methods [20, 36, 50] still assume source-domain pretraining, which directly conflicts with constraints on source-data access and is prone to overfitting in few-shot regimes. Therefore, it is necessary to adopt source-free test-time adaptation [34] that first selects the most informative layer for adaptation and then keeps the remaining backbone frozen while updating only a small subset of parameters.

To address these issues, we present Hierarchical Exemplar Representation Adaptation (HERA), as shown in Fig. 1, an efficient VFM-based segmentation framework that learns from limited labels and adapts to novel domains without source data retraining. We first propose Hierarchical Layer Selection (HLS) to estimate a per-episode Exemplar Transfer Risk (ETR) for each candidate layer from the few labeled supports and the forward activations, and choose the lowest risk layer as the working representation. We then confine test time updates to a small subset of parameters at this layer and keep the rest of the backbone frozen.

Even with a stable representation, token interaction maps, such as the self-attention maps used in ViTs, remain noisy under distribution shift. To mitigate this, Prior Guided Regularization (PGR) regularizes the attention on the selected representation with a lightweight spatial prior, strengthening locality and structural consistency while preserving global coverage. Finally, Pixelwise Adaptive Calibration (PAC) combines the selected representation with these prior guided attention maps and a query-image prototype-contrast map to calibrate pixel-level predictions, correcting residual artifacts along thin boundaries and in low-contrast regions. In summary, our contributions are:

- We present HERA, a source-free test-time adaptation framework for CD-FSS with VFMs that organizes adaptation as select, regularize, and calibrate, keeps the backbone frozen while fine-tuning less than 2.7% of parameters at test time in a lightweight manner.
- We introduce Hierarchical Layer Selection (HLS), driven by a data-dependent Exemplar Transfer Risk (ETR), which selects the lowest-risk layer per episode from a few labeled supports and cached forward activations, thereby localizing adaptation to a small parameter subset.
- We couple Prior-Guided Regularization (PGR) and Pixel-wise Adaptive Calibration (PAC) to regularize target-side structure and calibrate pixel-wise predictions, yielding a coherent hierarchy from representation to prediction.
- Extensive experiments on multiple CD-FSS benchmarks show consistent gains over prior methods, improving by over 4.1 mIoU, with ablations validating each stage and confirming strong parameter efficiency.

2. RELATED WORK

Few-Shot Semantic Segmentation (FSS) aims to predict pixel-level masks for novel classes using only a few labeled supports. Most existing methods fall into two families. Prototype-based approaches form class prototypes in feature space and classify queries by similarity [10, 27, 28, 31]. Affinity-based approaches compute dense correlations or attention between support and query features to propagate context [12, 37, 38, 41]. Subsequent works strengthen FSS through multi-scale context aggregation, prototype refinement, and mask-level regularization [4, 41, 47], yet most assume matched training and test distributions, leaving robustness to domain shift largely unexplored.

Cross-Domain Few-Shot Segmentation (CD-FSS) aims to generalize to novel classes in an unseen target domain using a few labeled support exemplars. Prior work is largely CNN-based and follows two lines. Source-side training performs domain generalization on source data before deployment [4, 13, 30, 46], which can be effective but requires continued access to source data and repeated retraining. Target domain supervised adaptation either mines correspondences across images or fine-tunes adapter modules using a few labeled supports [21, 39, 48, 55]. Despite progress, these pipelines are costly or source dependent, and convolutional inductive biases limit long range reasoning and robustness under distribution shift. In contrast, we adopt an effective VFM-based segmentation framework without source-data retraining.

Vision Foundation Models (VFMs) are largely ViT-based backbones pre-trained at scale with self-supervised, weakly-supervised, or semi-supervised objectives [7, 26, 40, 45]. Representative models include CLIP [42] for image-text alignment, MAE and EVA02 [14, 15, 19] for masked-image modeling, SAM [26] for promptable segmentation, and DINO [40, 45] for self-distillation with strong objectness cues. These models provide transferable

hierarchical features and often yield competitive segmentation with a frozen encoder. Specifically in DINOv3 [45], intermediate layers present coarse-to-fine semantics and attention that is locally coherent and globally aware, which suits support–query matching. Under distribution shift, however, layer utility varies across episodes and correspondence becomes unstable, so fixed layer choices or uniform fine-tuning are unreliable. We therefore adopt ViT-based VFMs with per-episode selection of a stable layer, followed by hierarchical regularization and calibration.

Test-Time Adaptation (TTA) adapts deployed models to target data using unlabeled test samples [24, 34, 52]. Common routes minimize entropy or consistency, update statistics, such as BN re-estimation, apply whitening or stylization, and perform contrastive or clustering-based alignment for segmentation [16, 25, 54]. Although deployment-friendly, they optimize surrogate losses on queries, require sizable trainable subsets or lengthy per-image updates, and are weakly coupled to the episodic nature of CD-FSS. In parallel, parameter-efficient fine-tuning (PEFT) updates a small fraction of weights via adapters, prompts, or low-rank modules [5, 6, 17, 22, 56], but for dense prediction it often targets single-level proxies and lacks episode-aware alignment. Our approach unifies TTA and PEFT by updating a small parameter subset at test time on a frozen backbone, guided by stable representation selection.

3. Method

3.1. Architecture Overview

Cross-domain few-shot segmentation (CD-FSS) follows an episodic K-shot protocol: given a support–query set $\mathcal{S} = \{(I_s^i, M_s^i)\}_{i=1}^K$ and a query image I_q, models trained on source domains are evaluated on target domains with disjoint label spaces. Leveraging vision foundation models (VFMs) is attractive. However, under distribution shift, per-episode alignment between support and query becomes unstable, causing errors to cascade from representation to prediction. In addition, VFM transferability varies across layers, and local interactions remain noisy.

We therefore propose Hierarchical Exemplar Representation Adaptation (HERA), a three-stage select-regularize-calibrate framework that adapts at test time with a frozen backbone. We first design Hierarchical Layer Selection (HLS) to select a stable representation layer ℓ^* by minimizing a data-dependent Exemplar Transfer Risk (ETR) computed per episode. Then Prior Guided Regularization (PGR) refines the self-attention at ℓ^* with entropy-gated Gaussian priors, strengthening locality and structural consistency while preserving global coverage. Finally, Pixel-wise Adaptive Calibration (PAC) calibrates pixel-wise predictions via lightweight residuals to recover thin structures and denoise low-contrast regions. Together, these stages form a hierarchical path from representation to prediction and transfer a few labeled supports into reliable guidance in new domains, yielding stable, well-calibrated predictions.

3.2. Hierarchical Layer Selection

Given a frozen ViT backbone f that produces layerwise features F^ℓ, we observe pronounced cross-layer transferability variance under distribution shift. We therefore make per-episode layer selection the primary stage of HERA. HLS minimizes the data-dependent episode-level ETR over candidate layers and outputs the selected representation F^{ℓ^*}, which conditions downstream modules.

3.2.1. Layerwise variability under shift

As illustrated in Fig. 3, early to mid layers 0-11 emphasize low-SNR edges and textures, whereas mid to high layers 12-23 yield class-agnostic objectness with sharper boundaries, with a consistent semantic shift around layers 11-12. Because the most informative layer varies across episodes and domains, any fixed choice is brittle. Episode-wise selection is thus crucial for robust support–query alignment. We therefore restrict routing to layers 12-23, where semantics consolidate while spatial detail is preserved. Single-layer routing in this band often performs well but is fragile on thin structures, occlusions, and clutter, as layers exhibit complementary failure modes. The optimal layer may fluctuate across episodes within a domain. To curb this instability, we augment the single-layer choice with a compact local-fusion candidate set centered at the best single layer and evaluate all candidates under a unified episode-level risk. This adds negligible cost and yields a more reliable routed representation for subsequent modules.

3.2.2. Per-Episode Layer Selection

Following Sec. 3.2.1, we estimate episode-level evidence at test time using leave-one-out and adapt only a small subset of parameters. Let the support set be $\mathcal{S} = \{(I_s^i, M_s^i)\}_{i=1}^K$. In the i-th iteration, (I_s^i, M_s^i) acts as the pseudo-query S_q^i and the remaining supports form $S^{(-i)}$. In the 1-shot setting, we synthesize two augmented views of the support for test time adaptation (TTA). Then we compute a prototype $\mathbf{P}_s^i$ from $S^{(-i)}$ and extract the pseudo-query feature $\mathbf{F}_q^i$ from I_s^i at a candidate layer ℓ.

We define Exemplar Transfer Risk (ETR) as one minus the average pseudo-query mIoU:

$$\mathcal{R}_{\text{layer}}(\ell) = 1 - \frac{1}{K}\sum_{i=1}^{K} \text{mIoU}\Big(\cos(\mathbf{P}_s^i, \mathbf{F}_q^i),\, M_q^i\Big), \quad (1)$$

$$\ell^\star = \arg\min_{\ell \in \mathcal{C}} \mathcal{R}_{\text{layer}}(\ell), \quad (2)$$

where $\mathcal{C}$ contains both single-layer and local-fusion candidates, and $\text{mIoU}(\cdot)$ compares the predicted mask with the

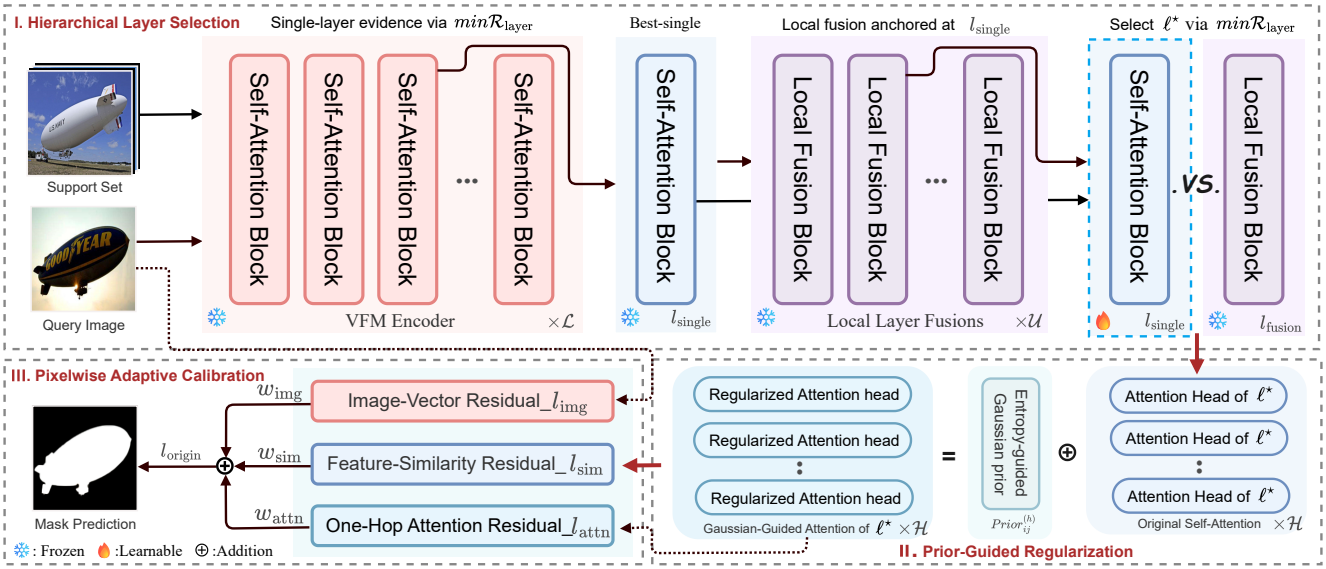


Figure 2. HERA architecture. Hierarchical Layer Selection (HLS) estimates the leave-one-out layer risk $\mathcal{R}_{\text{layer}}$ and routes each episode to $\ell^\star$, chosen from a single layer ℓ_{single} or a local-fusion candidate ℓ_{fusion}. At $\ell^\star$, Prior Guided Regularization (PGR) calibrates self-attention with entropy-gated Gaussian priors, and Pixelwise Adaptive Calibration (PAC) adds residual logits $\ell_{\text{sim}}, \ell_{\text{attn}}, \ell_{\text{img}}$ that are fused with the base logit in a select–regularize–calibrate hierarchy. $\mathcal{L}, \mathcal{U}$, and $\mathcal{H}$ are the numbers of layers, fusion candidates, and attention heads.

pseudo-query ground truth M_q^i, where M_q^i=M_s^i. The minimizer $\ell^\star$ is the routed layer used in subsequent stages.

With $\ell^\star$ fixed, we freeze the backbone and finetune only a small parameter set ϕ using the same leave-one-out construction, optimizing a binary segmentation loss:

$$\mathcal{L}_{\text{TTA}} = \frac{1}{K}\sum_{i=1}^{K} \text{BCE}\Big(\cos(\mathbf{P}_s^{i,\ell^\star}, \mathbf{F}_q^{i,\ell^\star}),\, M_q^i\Big), \quad (3)$$

where $\mathbf{P}_s^{i,\ell^\star}$ and $\mathbf{F}_q^{i,\ell^\star}$ are computed at the routed layer $\ell^\star$. To mitigate overfitting [1, 18], we fine-tune only the selected layer's MLP.FC for single-layer routing and fusion-MLP.FC for local-fusion routing and all remaining parameters are frozen. See Appendices 7 and 10 for more details.

3.2.3. Two-Stage Selection with Lightweight TTA

As shown in Algorithm 1, we first determine the best single layer ℓ_{single} via leave-one-out evidence and then form a compact set of local-fusion candidates $U \subseteq \mathcal{L} = \{12, \ldots, 23\}$ anchored at ℓ_{single}. Empirically, we observe that the last ViT block tends to provide more category-consistent and globally coherent features with strong objectness. We therefore include ℓ_{23} in each local-fusion candidate as a global context anchor to compensate for occlusion and fragmented shapes. All candidates are evaluated using the same episode-level risk ETR in Eq. (1).

For any candidate U, let $r_\ell = \mathcal{R}_{\text{layer}}(\ell)$ denote the single-layer selection risk. Define the distance $d(\ell, \ell_{23}) = |\ell - \ell_{23}|$ and compute fusion weights and the fused feature as:

$$w_\ell = \frac{\exp\big(-\beta r_\ell - \text{dist}(\ell, \ell_{23})/\tau\big)}{\sum_{j\in U}\exp\big(-\beta r_j - \text{dist}(j, \ell_{23})/\tau\big)}, \quad (4)$$

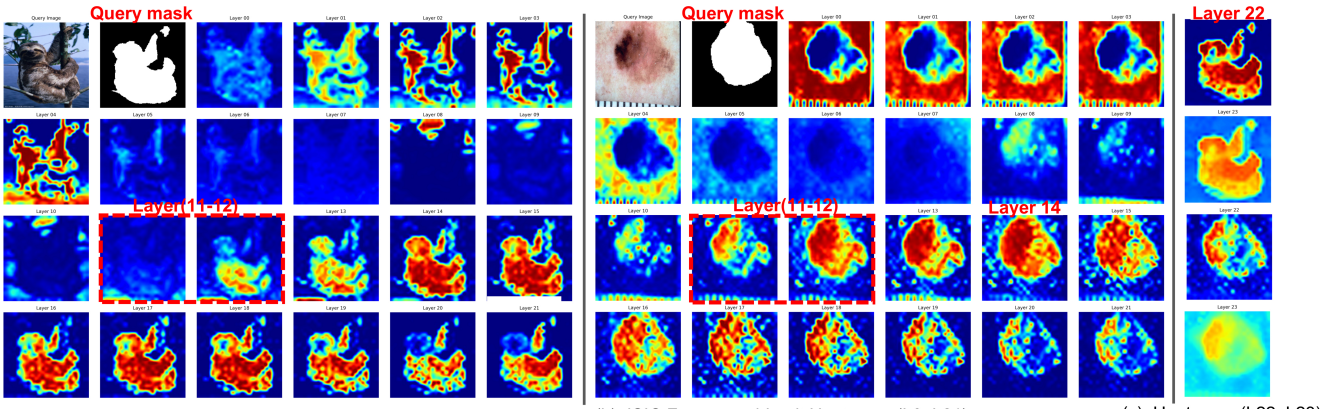
$$F^U = \sum_{\ell\in U} w_\ell F^\ell, \quad (5)$$

where $\beta > 0$ controls reliance on the evidence r_ℓ and $\tau > 0$ is a locality bandwidth that favors deeper, semantically aggregated layers. As $\tau \to \infty$, the locality term vanishes; as $\beta \to \infty$, the weights collapse to $\arg\min_{\ell\in U} r_\ell$, approaching the single-layer case. When evidence spreads across adjacent layers, a moderate τ trades off evidence and aggregation, reducing routing instability.

Fixing the routed layer at $\ell^\star$ mitigates the layer-wise component of episode-wise matching risk, thereby providing a more stable, episode-conditioned representation for downstream adaptation stages.

3.3. Prior Guided Regularization

With the routed layer fixed by HLS, the representation provides stable global semantics, yet head-specific self-attention remains noisy under shift. Because DINO features provide mainly layer-level guidance, head-level maps show spurious long-range links, insufficient coverage of nearby neighborhoods, thin boundaries, and strong cross-head heterogeneity. A uniform, head-agnostic prior is therefore inadequate. We calibrate attention per head using a query-centered Gaussian prior whose bandwidth is set by an entropy gate derived from the head's attention. Local and con-



(a). FSS Foreground-Logit Heatmaps (L0–L21) (b). ISIC Foreground-Logit Heatmaps (L0–L21) (c). Heatmaps (L22–L23)

Figure 3. Layerwise variability in VFM features (DINOv3 example). Foreground-logit heatmaps from ViT layers 0-23 for two episodes from distinct target domains. A semantic shift emerges around layers 11-12, yet the optimal layer for mask prediction differs across episodes, indicating episode- and domain-dependent variability in the best layer and motivating per-episode layer routing.

Algorithm 1 Hierarchical Layer Selection (HLS)

Require: K-shot support set $\mathcal{S}$, candidate layers $\mathcal{L} = \{12, \ldots, 23\}$ and VFM backbone f.
Select trainable parameters
for $n = 1$ to $K-1$ **do** ◊ ***Episode-level routing evidence (n-shot): see Sec. 3.2***
  ▷ ***Stage I: Single-layer evidence***
    Assemble the n-shot episode S_n; extract per-layer features $\{F^\ell\}_{\ell \in \mathcal{L}}$ with f
    Compute $\mathcal{R}_{\text{layer}}(\ell)$ on S_n and obtain ℓ_{single} *single risk, see Eq. (1)*
  ▷ ***Stage II: Local fusion around ℓ_{single}***
    Construct a local fusion pool $\mathcal{U}(\ell_{\text{single}})$
    For each $U \in \mathcal{U}$, compute fusion weights w_ℓ and fused feature F^U *see Eqs. (4) and (5)*
    Choose $\ell^\star = \arg\min_{\{\ell_{\text{single}}\} \cup \mathcal{U}(\ell_{\text{single}})} \mathcal{R}_{\text{layer}}(\ell)$ and fix $F^{\ell^\star}$ *unified risk, see Eq. (2)*
  for $i = 1$ to K **do** ◊ ***Pseudo–query cross-evaluation***
    Form $\langle S^{(-i)}, (I_s^i, M_s^i) \rangle$ at fixed $F^\star$
    Compute $\mathcal{L}_{\text{TTA}}^{(i,n)}$ over all n-support combinations *see Eq. (3)*
  end for
  Update ϕ by back-propagating the average loss $\mathcal{L}_{\text{TTA}}^{(n)}$; keep f frozen ◊ ***Parameter-efficient TTA***
end for

fident heads receive a sharper prior, whereas globally dispersed heads receive a more diffuse prior. This head-wise, entropy-gated calibration enforces locality while preserving necessary global coverage as shown in Fig. 4

Head-wise Gaussian Prior with Entropy Gating. Attention heads in ViTs exhibit specialization in spatial scope and semantics [29, 43, 44]. We therefore impose a head-wise Gaussian prior and set its bandwidth by an entropy gate, yielding a sharper prior for local, confident heads and a more diffuse prior for globally dispersed ones:

$$\phi(p_j;\, p_i, \sigma) = \exp\Big(- \frac{\|p_j - p_i\|^2}{2\sigma^2} \Big), \tag{6}$$

with two fixed bandwidth hyperparameters, $\sigma_{\text{loc}} < \sigma_{\text{glo}}$, capturing local aggregation and global coverage.

Let $\bar{H}_q^{(h)}$ denote the mean row entropy of the $QK^\top$ logits at head h, indicating global dispersion, and $\bar{H}_k^{(h)}$ that of

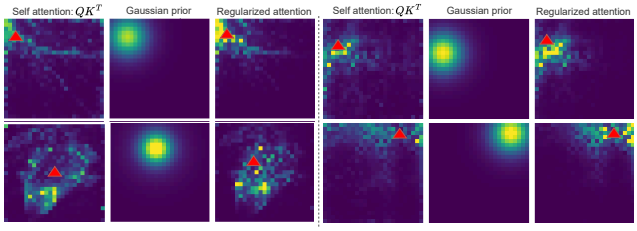


Figure 4. Prior Guided Regularization (PGR). Per-head Gaussian priors, gated by entropy, are injected into $QK^\top$ logits to calibrate self-attention at $\ell^\star$, locality is strengthened, far-field spurious peaks are suppressed, while preserving global coverage.

$KK^\top$, indicating local stability. Using a logistic gate $g(\cdot)$ with temperature $\alpha > 0$:

$$\begin{aligned} \gamma_h &= g\big(\alpha(\bar{H}_q^{(h)} - \bar{H}_k^{(h)})\big), \\ \sigma_h &= (1 - \gamma_h)\, \sigma_{\text{glo}} + \gamma_h\, \sigma_{\text{loc}}, \end{aligned} \tag{7}$$

Table 1. Quantitative comparison on the CD-FSS benchmark. All compared methods, except HERA, are trained on the Pascal VOC [11] source domain and then evaluated on four targets, whereas HERA is source-free and performs no source- or target-side training (test-time adaptation only). Best and second-best results are shown in **bold** and underlined. The † indicates results reproduced by us, and ‡ indicates a ViT-base backbone. In the Training (S/T) column, ✓/✓ denotes source-side pretraining and supervised target retraining; ✓/× denotes source-side pretraining only (no target retraining); ∅ denotes source-free, no target-side domain retraining, test-time adaptation only.

Methods	Publication	Training (S/T)	DeepGlobe		ISIC2018		Chest X-ray		FSS-1000		mIoU	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
PGNet [58]	ICCV 2019	✓/×	10.7	12.4	21.9	21.3	34.0	23.0	62.4	62.7	32.2	31.1
PANet [53]	ICCV 2019	✓/×	36.6	45.4	25.3	34.0	57.8	69.3	69.2	71.7	47.2	55.1
CaNet [59]	CVPR 2019	✓/×	22.3	23.1	25.2	28.2	28.4	28.6	70.7	72.0	36.6	38.0
RPMMs [57]	ECCV 2020	✓/×	13.0	13.5	18.0	20.0	30.1	30.8	65.1	67.1	31.6	32.9
PFENet [47]	TPAMI 2020	✓/×	16.9	18.0	23.5	23.8	27.2	27.6	70.9	70.5	34.6	35.0
RePRI [1]	CVPR 2021	✓/×	25.0	27.4	23.3	26.2	65.1	65.5	71.0	74.2	46.1	48.3
HSNet [38]	ICCV 2021	✓/×	29.7	35.1	31.2	35.1	51.9	54.4	77.5	81.0	47.6	51.4
SSP† [12]	ECCV 2022	✓/×	40.5	49.6	35.5	48.2	74.2	74.5	79.0	80.2	57.3	63.1
DATO [33]	CVPR 2025	✓/×	**51.1**	59.3	**68.8**	70.3	79.6	81.1	81.8	84.6	**70.3**	73.8
PATNet [30]	ECCV 2022	✓/✓	37.9	43.0	41.2	53.6	66.6	70.2	78.6	81.2	56.1	62.0
PMNet [3]	WACV 2024	✓/✓	37.1	41.6	51.2	54.5	70.4	74.0	**84.6**	86.3	60.8	64.1
ABCDFSS [21]	CVPR 2024	✓/✓	42.6	49.0	45.7	53.3	79.8	81.4	74.6	76.2	60.7	65.0
APSeg‡ [20]	CVPR 2024	✓/✓	35.9	40.0	45.4	54.0	84.1	84.5	79.7	81.9	61.3	65.1
DR-Adapter [46]	CVPR 2024	✓/✓	41.3	50.1	40.8	48.9	82.4	82.3	79.1	80.4	60.9	65.4
APM [48]	NeurIPS 2024	✓/✓	40.9	44.9	41.7	51.2	78.3	82.8	79.3	81.9	60.0	65.2
IFA [39]	CVPR 2024	✓/✓	50.6	58.8	66.3	69.8	74.0	74.6	80.1	82.4	67.8	71.4
TVGTANet [35]	ACM MM2025	✓/✓	42.0	50.7	47.2	58.8	84.6	87.3	78.3	81.4	63.0	69.5
SDRC‡ [50]	ICML 2025	✓/✓	43.2	46.8	46.6	55.0	82.9	84.8	80.3	82.6	63.2	67.3
DFN [49]	ICML 2025	✓/✓	39.5	47.7	50.4	58.5	83.2	87.1	83.0	85.7	64.0	69.8
LoEC‡ [36]	CVPR 2025	✓/✓	42.1	51.5	52.9	62.4	83.9	84.1	81.1	83.7	65.0	70.4
HERA‡ (DINOv2)	—	∅	41.2	57.8	55.6	68.7	83.2	86.9	70.2	80.3	62.6	73.4
HERA‡ (DINOv3)	—	∅	44.6	**63.4**	61.2	**73.6**	**85.8**	**87.9**	81.6	**86.7**	68.3	**77.9**

where heads with stronger locality, indicating larger $\bar{H}_q^{(h)} - \bar{H}_k^{(h)}$, receive a sharper prior, and globally dispersed heads receive a more diffuse prior.

3.4. Pixelwise Adaptive Calibration

With the layer and attention stages in place, pixelwise decisions remain unstable under shift, leaving residual errors along thin boundaries and in low-contrast regions. PAC explicitly performs pixelwise calibration by fusing three lightweight cues computed from F^{ℓ^*} and the refined attention, namely feature similarity, one-hop attention propagation, and image appearance. This fusion corrects residual artifacts and yields consistent masks:

$$\ell_{\text{final}}(x) = \ell_0(x) + w_{\text{sim}}\ell_{\text{sim}}(x) + w_{\text{attn}}\ell_{\text{attn}}(x) + w_{\text{img}}\ell_{\text{img}}(x) \quad (8)$$

where $\ell_0(x)$ is the base logit from the selected representation and $w_\cdot$ are fixed scalar weights. A single-step refine-vote gate applies residuals only when the estimated gain is positive, adding negligible overhead. Taken together, the three stages realize a hierarchical select-regularize-calibrate pipeline that adapts at test time with a frozen backbone. See Appendix 5 for more details.

4. Experiments

Evaluation Datasets and Metrics. We evaluate in a source-free test-time adaptation setting without access to source data. We evaluate on four public target datasets: FSS-1000 [32], a natural image corpus spanning 1,000 object classes; DeepGlobe [9], a satellite land cover dataset with seven categories and pronounced texture and scale shifts; ISIC2018 [8, 51], a dermoscopic skin lesion dataset with low-contrast and irregular boundaries; and Chest X-ray [2, 23], a medical radiograph dataset with substantial grayscale and structural variation. We follow the standard episodic K-shot protocol and report the mean IoU for 1-shot and 5-shot regimes. To ensure comparability, all methods use identical episode sampling, a shared preprocessing pipeline, and a unified input resolution of 400×400.

Implementation Details. We use SSP [12] as the few-shot head and run HERA on a ViT backbone (default DI-

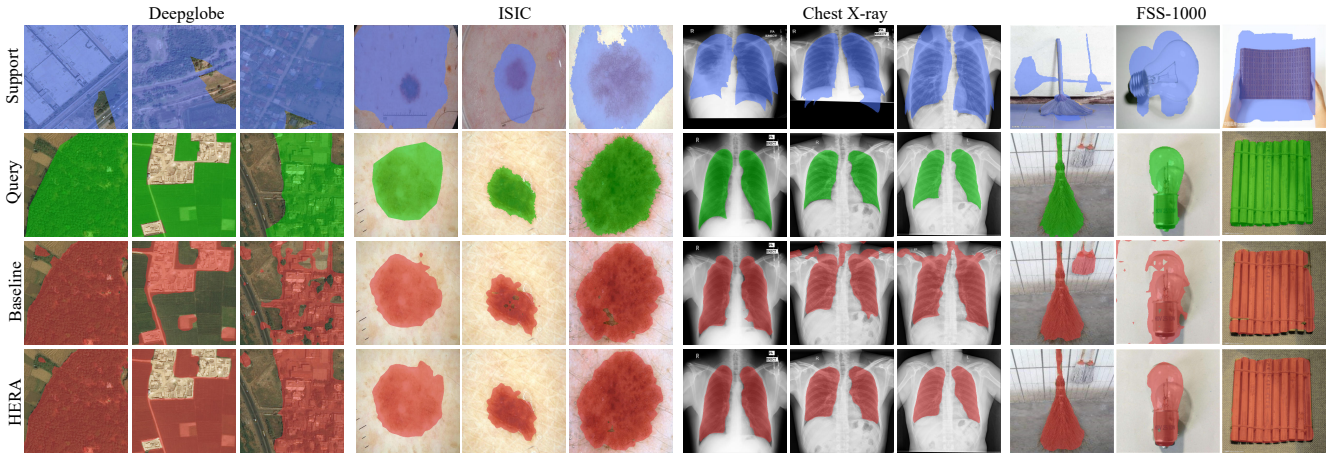


Figure 5. Qualitative results on the Chest X-ray, ISIC, FSS-1000, and Deepglobe datasets under the 1-shot setting. The prediction and ground truth of query images are in red and green, respectively. Best viewed in color.

NOv3 [45]). Under test-time adaptation (TTA), each target episode proceeds as follows: (i) HLS selects the routed layer; and (ii) we form leave-one-out splits of the K supports and minimize the loss in (3) on these splits, performing $K-1$ lightweight updates. We optimize with Adam (learning rate 1.3×10^{-3}, $\beta_1 = 0.9$, $\beta_2 = 0.999$). Only a small parameter subset is trainable: the single-layer variant updates that layer's mlp.fc, and the local-fusion variant updates the fusion-mlp.fc and all other weights remain frozen. In the 1-shot setting, we synthesize two augmented views from the single support using a soft copy–paste scheme to stabilize TTA. All experiments run on a single NVIDIA A100, with peak VRAM of 4.2 GB (5% of capacity). Since fewer than 2.7% of parameters are updated, both compute and memory overheads are negligible.

4.1. Comparison Experiments

Our HERA unifies VFM-based adaptation for CD-FSS into a source-free, no-retraining test-time pipeline that is fundamentally different from the training paradigms of the methods reported in Table 1. Under an identical evaluation protocol with the same episode sampling, preprocessing, and 400×400 input, HERA with DINOv3 attains 68.3/77.9 mIoU in the 1-shot/5-shot settings. It outperforms LoEC [36] by +3.3/+7.5, SDRC [50] by +5.1/+10.6, and the no-retraining baseline SSP by +11.0/+14.8 mIoU. Even compared with the CNN-based DATO [33], which requires source-domain training, HERA leads by +4.1 in the 5-shot setting and exceeds the 5-shot performance of all other CNN-based methods. At the same time, HERA remains source-free, which better satisfies privacy and engineering constraints and simplifies deployment.

In the 1-shot setting, HERA also provides practical advantages in both performance and efficiency as shown in Fig. 5. It surpasses TVGTANet [35] and DFN [49] by +5.3 and +4.3 mIoU, respectively, even though these methods require training on both source and target domains. The proposed denoising and boundary calibration modules are specifically designed for HERA and yield natural, well-defined boundaries. On Chest X-ray, HERA achieves the best results with 85.8/87.9 mIoU, indicating the combination of selection–regularization–calibration is particularly effective for low-contrast, thin-boundary clinical cases.

Overall, HERA delivers state-of-the-art performance in the 5-shot setting without any source or target training and remains strong in the 1-shot setting, while updating only $< 2.7\%$ of its parameters. This yields a favorable trade-off between accuracy and cost, and makes deployment across institutions and devices easier and less constrained. Analyses with other VFMs are provided in Appendix 8.

Table 2. HERA component ablation (5-shot mIoU↑, frozen VFM, 400×400). HLS/PGR/PAC as defined; 'Δ Avg.' = gain over SSP.

Method	DeepGlobe	ISIC	Chest X-ray	FSS-1000	5-shot	Δ Avg.
Baseline	49.6	48.2	74.5	80.2	63.1	0.0
+ HLS	61.7	71.4	87.7	86.0	76.7	+13.6
+ HLS + PGR	62.6	72.0	88.0	86.5	77.3	+14.2
+ HLS + PAC	62.1	71.6	**88.3**	86.6	77.2	+14.1
+ HLS + PGR + PAC	**63.4**	**73.6**	87.9	**86.7**	**77.9**	**+14.8**

Table 3. PAC branch ablation on HLS+PGR (5-shot mIoU↑, frozen VFM). 'Δ Avg.' = gain vs. HLS+PGR.

Variant	5-shot	Δ Avg.
Baseline + HLS + PGR	77.27	–
+ l-sim	77.57	+0.30
+ l-attn	77.49	+0.22
+ l-img	77.45	+0.18
+ l-sim + l-attn + l-img	**77.91**	**+0.64**

Table 4. Layer selection ablation with a frozen backbone (5-shot mIoU↑). Rule lists the per episode selector and notation includes $\mathbf{g}_\ell$ gradient w.r.t. features of layer ℓ; S_{sem}, S_{str}, C normalized semantic, structure, and complexity scores; and $\text{mIoU}_{\text{sup}}(\ell)$ support only mIoU.

Method	Rule	DeepGlobe	ISIC	Chest X-ray	FSS-1000	Avg.	Δ
Static-Max	$\arg\max_\ell \left(\alpha\, S_{\text{sem}}(\ell) + \beta\, S_{\text{str}}(\ell) + \gamma\, C(\ell)\right)$	58.8	<u>67.2</u>	80.1	81.4	71.9	0.0
Grad-Max	$\arg\max_\ell \|\mathbf{g}_\ell\|$	60.2	62.0	85.4	84.7	73.1	+1.2
GradΔ-Max	$\arg\max_\ell \|\mathbf{g}_\ell - \mathbf{g}_{\ell-1}\|$	<u>60.4</u>	61.7	<u>86.0</u>	<u>84.8</u>	<u>73.2</u>	<u>+1.3</u>
HLS (ETR)	$\arg\min_\ell \left(1 - \text{mIoU}_{\text{sup}}(\ell)\right)$	**61.7**	**71.4**	**87.7**	**86.0**	**76.7**	**+4.8**

4.2. Ablation Studies

Component ablation. We ablate HERA in the 5-shot setting with a frozen VFM backbone as shown in Table 2. The SSP baseline averages 63.1 mIoU, adding HLS raises the mean to 76.7 mIoU, which provides the dominant gain. With HLS fixed, attaching PGR or PAC raises the average to 77.3 and 77.2 mIoU, respectively. These improvements are positive and stable across all four datasets, with per-dataset gains of +0.2 to +0.9 mIoU on average. The full-stack HLS+PGR+PAC reaches 77.9 mIoU, a gain of total +14.8 mIoU over SSP. Qualitatively, PGR suppresses spurious far-field correlations and strengthens local coherence, whereas PAC calibrates pixels in low-contrast, thin-boundary regions, yielding cleaner borders and less leakage. These effects also reduce failures in the 1-shot regime, and the marginal cost of adding these modules is minimal. We update $< 2.7\%$ of the parameters, and the additional computational overhead is negligible. Overall, HLS aligns the routed representation, while PGR and PAC provide robust, interpretable structural refinements, yielding a cumulative top-down select-regularize-calibrate effect.

Interaction of PGR and PAC Components. With HLS at 76.7 mIoU, adding PGR raises the mean to 77.3 mIoU, adding PAC yields 77.2 mIoU, and enabling both reaches 77.9 mIoU. The corresponding gains over HLS are 0.6, 0.5, and 1.2 mIoU, see Table 2, which is slightly above 1.1 and indicates complementarity rather than redundancy. Decomposing PAC in Table 3 relative to HLS+PGR, the similarity residual ℓ_{sim}, the one-hop attention prior ℓ_{attn}, and the image cue ℓ_{img} add 0.30, 0.22, and 0.18 mIoU, respectively, and using all three reaches 77.91 mIoU, an additional 0.64 mIoU over HLS+PGR. PGR regularizes attention at the representation level, while PAC calibrates predictions at the pixel level, thereby making the two modules complementary and consistently effective.

Effect of Per-Episode Layer Selection (HLS). We compare per-episode selectors for test-time adaptation with a frozen backbone, as shown in Table 4. Our criterion-based HLS routes each episode to the layer that maximizes support-only mIoU, $\arg\min_\ell\left(1 - \text{mIoU}_{\text{sup}}(\ell)\right)$, and provides task-aligned, episode-aware feedback without extra parameters or surrogate losses. In contrast, Static-Max scores feature quality using semantic, structural, and complexity heuristics rather than task fit. Gradient proxies, including Grad-Max and GradΔ-Max, rank layers by gradient magnitude or change. These proxies tend to favor the final blocks in ViTs because of residual connections and normalization, and they correlate weakly with semantic alignment. HLS attains the best average performance of 76.7 mIoU, surpassing Static-Max by +4.8 mIoU and the strongest gradient proxy by +3.5 mIoU. These results justify per-episode layer selection as the primary performance driver and the natural entry point of HERA. The gap is larger on other VFM backbones, for example, DINOv2 [40].

Adaptation Cost in New Target Domains. Many retraining-based and domain-generalization methods require tens to hundreds of GPU-hours of computation on source data prior to deployment. In contrast, HERA follows a source-free, no-retraining test-time adaptation paradigm. For a new target domain, we run a single adaptation phase driven by the available supports and then reuse the adapted model for all remaining queries. Using a DINOv3 backbone in the 1-shot setting, the per-episode costs are 0.202 s for hierarchical layer selection, 0.280 s for the lightweight parameter update, and 0.243 s for inference, with only 8.39M trainable parameters (2.69%). Each stage, therefore, adds only a small fraction of latency and memory relative to standard inference. This design allows HERA to quickly specialize a frozen VFM to a new domain while still delivering strong cross-domain performance.

5. Conclusion

We identify layer-wise transferability variation and noisy head-level interactions under domain shift, rather than limited representational capacity, as the key bottlenecks when applying VFMs to CD-FSS. We introduce Hierarchical Exemplar Representation Adaptation (HERA), a three-stage select-regularize-calibrate VFM-based segmentation framework that learns effectively from limited labels and adapts to novel domains without source-data retraining. Across domains and backbones, HERA improves mIoU by over 4.1 while updating 2.69% of its parameters, providing a low-overhead and practical way to deploy VFMs in CD-FSS.

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